

NWC 883 / NETWORK-BASED COMPUTING FOR HPC AND AI

PARALLEL AGGREGATED TREES

PAT: A NEW ALGORITHM FOR ALL-GATHER AND REDUCE-SCATTER OPERATIONS AT SCALE

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WHAT IS PAT?

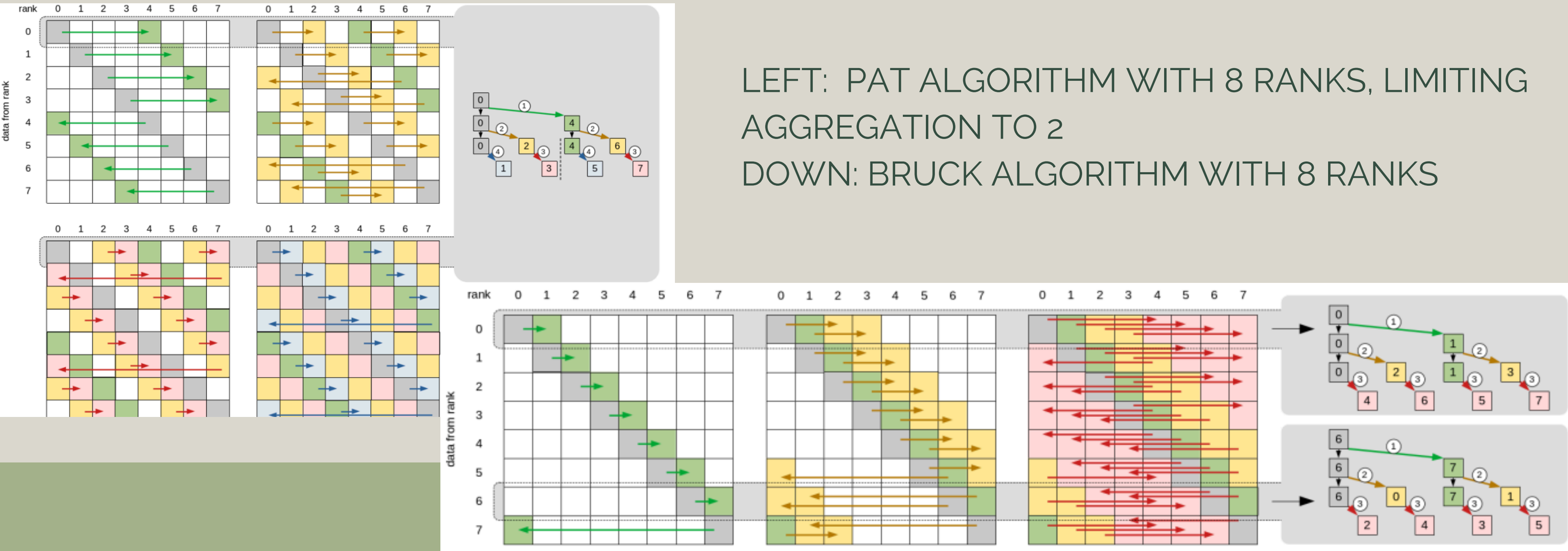
- ALGORITHM TO IMPLEMENT ALL-GATHER AND REDUCE-SCATTER OPERATIONS
- DERIVED FROM THE BRUCK ALGORITHM FOR ALL-GATHER
- IMPLEMENTED IN NCCL 2.23

WHY IS PAT?

- WORKS ON ANY NUMBER OF RANKS
- LOGARITHMIC FOR SMALL SIZE OPERATIONS
- LOGARITHMIC + LINEAR FOR LARGER OPERATIONS
- USES A LOGARITHMIC NUMBER OF INTERNAL BUFFERS
- MINIMAL COMMUNICATION VOLUME BETWEEN DISTANT RANKS

PAT VS BRUCK

- “IF THE MAXIMUM AMOUNT OF DATA WE SEND TO A GIVEN RANK FITS INSIDE THE INTERMEDIATE BUFFER, IT (PAT) WILL LOOK EXACTLY LIKE THE REVERSED-DIMENSION BRUCK ALGORITHM.”



ALGORITHM

- INITIATE COMMUNICATION
- INITIATE BUFFERS
- CALCULATE NUMBER OF LOGARITHMIC STEPS
- LOGARITHMIC PHASE:
 1. $\text{SEND_PARTNER} = (\text{RANK} + 2^{\text{STEP}}) \% P;$ $\text{RECV_PARTNER} = (\text{RANK} - 2^{\text{STEP}} + P) \% P;$
 2. AFTER ALL STEPS COMPLETED, SWITCH TO LINEAR PHASE
- LINEAR PHASE:
 1. CALCULATE APPROPRIATE SEND AND RECEIVE COUNT (FOR NON POWERS OF 2)
 2. $\text{SEND_INDEX} = (I * \text{INDEX} + \text{RANK}) \% P;$ $\text{RECV_INDEX} = (\text{RANK} - (I * \text{INDEX}) + (P * I)) \% P;$
(WHERE INDEX IS THE DISTANCE IN THE LAST STEP OF LOGARITHMIC PHASE)
- REORDER BUFFERS

MODIFICATIONS WE EXPLORED

COMPRESSION:

- TRIED COMPRESSING THE DATA OF EACH RANK BEFORE COMMUNICATION.
- AS BUFFER SIZE IS CONSTRAINED, COMPRESSION MAY ALLOW US TO INCREASE THE NUMBER OF LOGARITHMIC STEPS WE TAKE, THUS SAVING TIME.
- TRADEOFFS BETWEEN TIME SAVED DURING COMMUNICATION, REDUCTION IN COMMUNICATION VOLUME, AND TIME AND RESOURCES REQUIRED FOR ADDITIONAL COMPUTE.

DYNAMIC BUFFER SIZE SETTING:

- QUERY THE SYSTEM TO GET THE MAXIMUM BUFFER SIZE THAT CAN BE SAFELY USED.
- LARGER BUFFER SIZE = MORE LOGARITHMIC STEPS = TIME SAVED

IMPLEMENTATION

GITHUB LINK: [PAT IMPLEMENTATION](#)

MAIN FEATURES:

- IMPLEMENTED IN C
- USES THE OPENMPI IMPLEMENTATION OF MPI FOR COMMUNICATION
- CURRENTLY IMPLEMENTED FOR INTRA NODE COMMUNICATION

TESTING:

- TESTED FOR UPTO 32 RANKS ON A SINGLE LAPTOP
- TESTED ALL-GATHER WITH EACH RANK HAVING UPTO 1024 INTEGER DATA CHUNKS
- TESTED ALLOCATED BUFFER SIZE UPTO 32 MB PER RANK (8192 INTEGERS)

KEY ISSUES WE RAN INTO

IMPLEMENTATION:

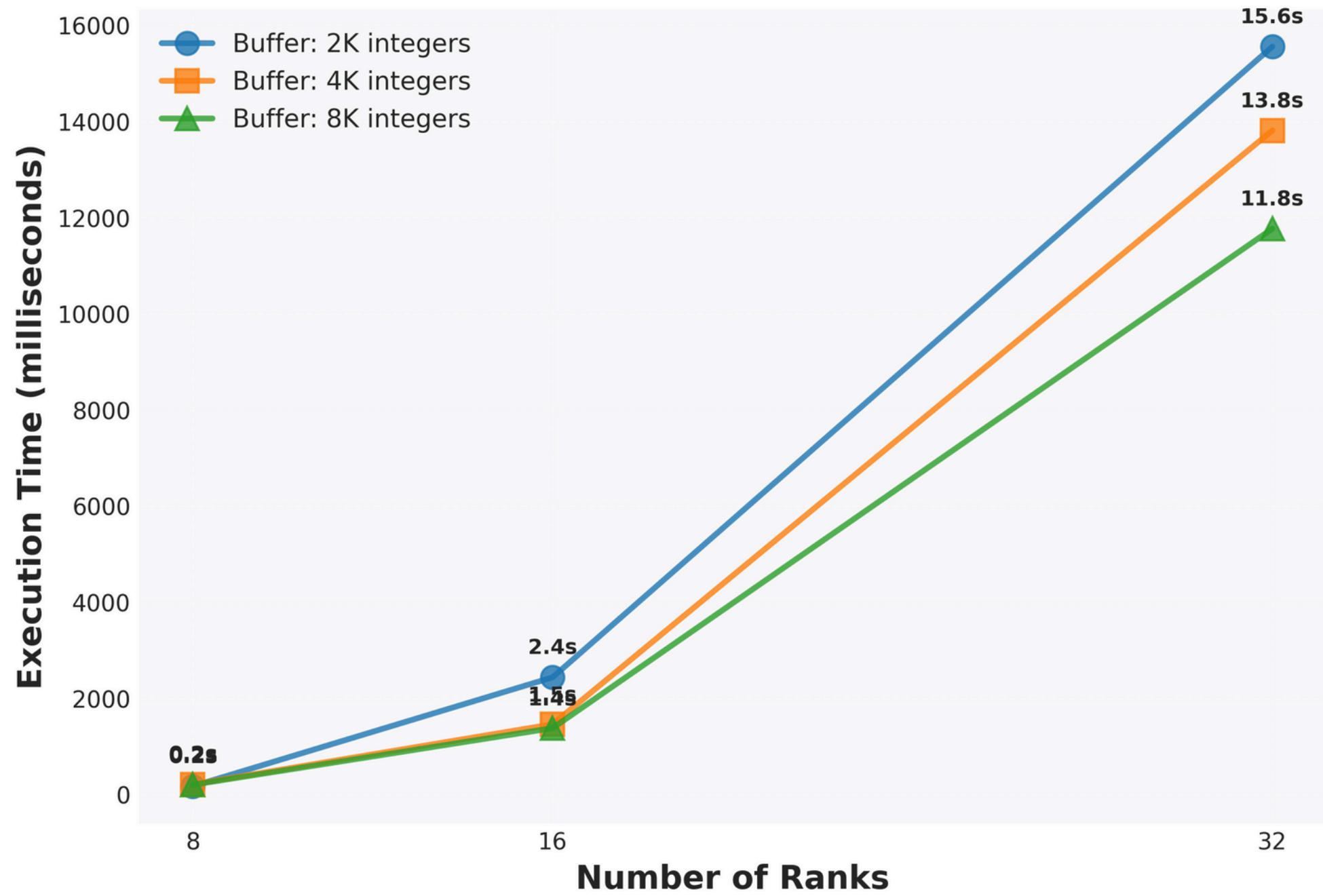
- CASES WHEN NUMBER OF RANKS WERE NOT A POWER OF 2
- SWITCHING FROM THE LOGARITHMIC TO LINEAR PHASE
- CORRECT REORDERING OF BUFFERS

OPTIMIZATION:

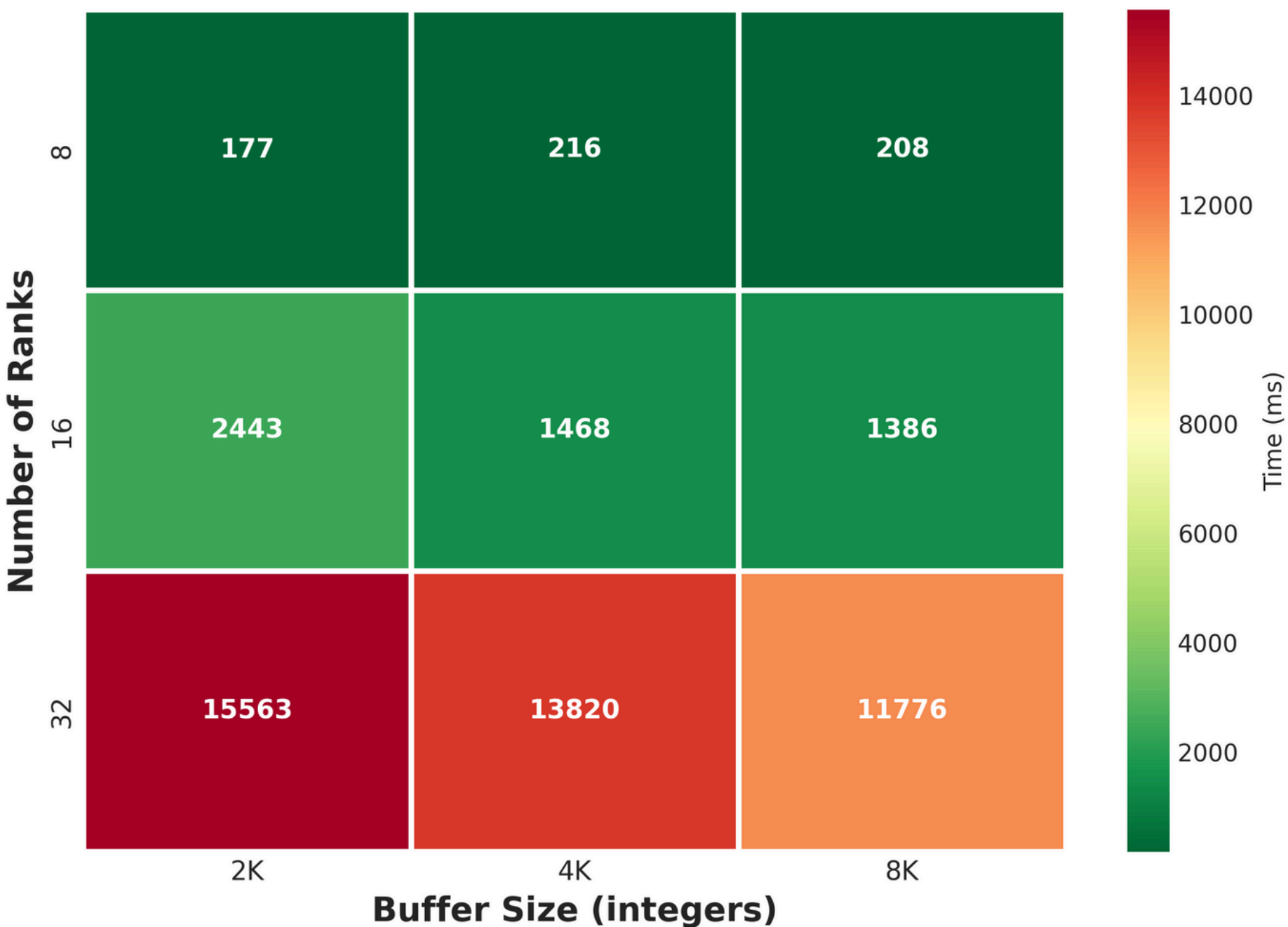
- LONG WINDED / UNNECESSARY APPROACHES
- EXCESSIVE NUMBER OF COMMUNICATION BARRIERS
- (YET TO BE COMPLETELY FIXED) EXCESSIVE NUMBER OF MALLOCs AND MEMORY COPIES

BENCHMARKING

Scalability Analysis: Execution Time vs Process Count



Performance Matrix: Execution Time Heatmap



COMPRESSION AND ITS TRADEOFFS

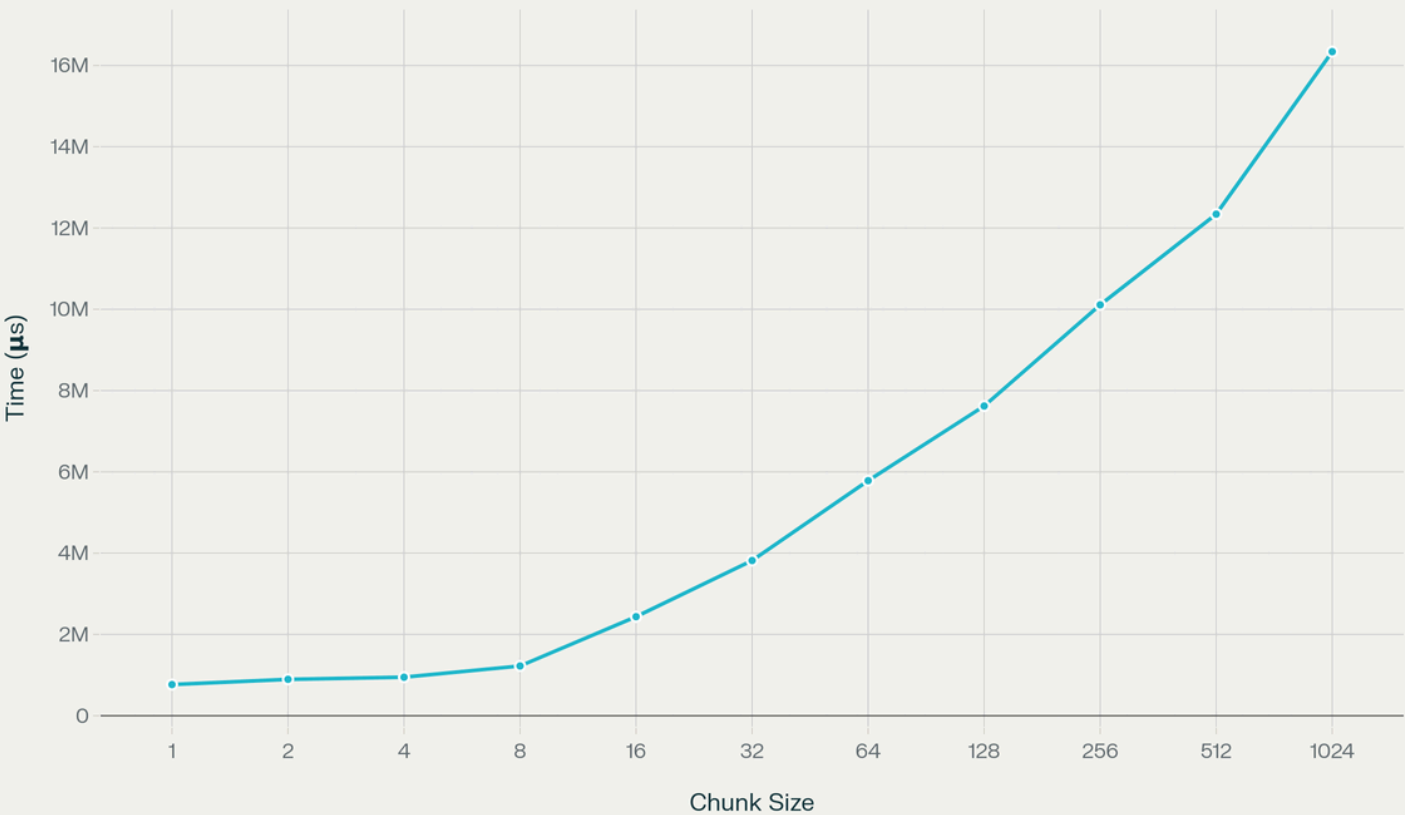
SPARSE COMPRESSION:

- USED WHEN A LARGE PORTION OF YOUR DATA IS FILLED WITH ZEROES.
- COMPRESSION FACTOR: $\frac{50}{(100 - n)} X$, WHERE N IS THE PERCENTAGE OF ZEROES (ARRAY).

RUN LENGTH ENCODING:

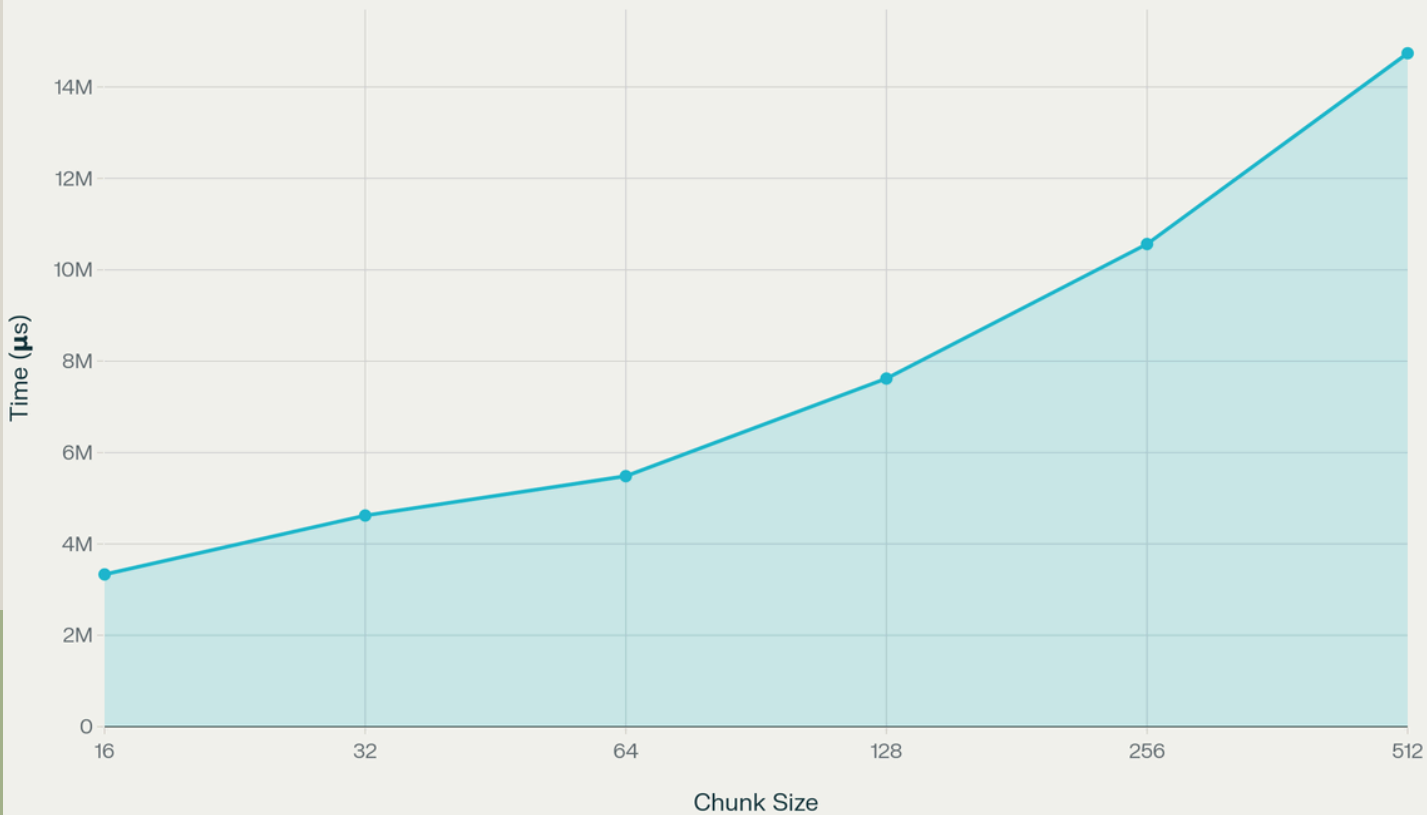
- USED WHEN VALUES ARE MANY CONSECUTIVE VALUES ARE REPEATED.
- CAN BE USED TO COMPRESS IMAGES WITH LARGE MONOCHROME SEGMENTS OR VIDEOS WITH STATIC BGS.

PAT AllGather Performance



LEFT:
(BUFFER SIZE / CHUNK
SIZE) = 8 (CONSTANT)
RIGHT:
BUFFER SIZE = 1024
(CONSTANT)

PAT AllGather: Chunk vs Exec Time



COMPARISON TO MPI

UNFORTUNATELY, OUR IMPLEMENTATION IS STILL MUCH SLOWER THAN THE INBUILT METHODS PROVIDED BY MPI

POTENTIAL REASONS:

- LARGER SYNCHRONIZATION OVERHEAD
- TOO MANY MALLOCs AND FREES (EXPENSIVE OPERATIONS)
- TOO MANY MEMCPYS (EXPENSIVE AS WELL)
- POTENTIAL IMPLEMENTATION ERRORS

FUTURE WORK/ IMPROVEMENTS

- OPTIMIZE THE CURRENT IMPLEMENTATION BY:
 1. REDUCING SYNCHRONIZATION OVERHEADS
 2. REDUCING NUMBER OF MALLOCs AND MEMCPYS
- IMPLEMENT REDUCE-SCATTER AND THUS ALL-REDUCE USING PAT.
- IMPLEMENT IT USING NCCL.
- IMPLEMENT IT FOR MULTIPLE NODES.
- EXPLORE MORE OPTIMIZED COMPRESSION METHODS, AND EXPLORING LOSSY METHODS OF COMPRESSION FOR GREATER SPEED

THE END

THANK YOU
FOR LISTENING

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